

**I pivoted FY2020, FY2021
into columns on one row**
**Zero FY2020 sales nearly
broke the growth math**

ATLIQ HARDWARE GDB0041

MYSQL | Vishnu Ram Sachin | Data Analyst



Problem Statement

DAY 11 / 15

06/08/26

- Yesterday : Ranked products with RANK() OVER (
- PARTITION BY division)
- Leadership wants markets that grew **FY2020 VS FY2021**

The Trap ❌

- GROUP BY market , fiscal_year gives two seperate rows per market

Two rows per market , not one

can't subtract FY2021- FY2020 across rows



Requirment

DAY 11 / 15

06/08/26

FY2020 in its own column



FY2021 in its own column



Split the year first ,
then place side by side per market

The Fix

Use Window Function

```
SUM (CASE WHEN fiscal_year = 2020  
THEN gross_sales ELSE 0 END ) as fy_2020_sales
```



A Quick Analogy

DAY 11 / 15

06/08/26

Bank Statement

Every Transaction

months mixed in one long list
(GROUP BY month)

Split by month

Jan , Feb , March each get a column
(CASE WHEN pivot)

One row per account

every month now sits side by side
(single row output)

Compare two columns

This month vs last month = change
(Growth %)



```
WITH gross_sales_report AS (  
SELECT  
dc.customer_code , dc.customer ,  
    dc.channel, dc.market, dd.fiscal_year,  
fsm.sold_quantity * fgp.gross_price AS gross_sales  
FROM fact_sales_monthly fsm  
JOIN dim_customer dc  
ON dc.customer_code = fsm.customer_code  
JOIN fact_gross_price fgp  
ON fgp.product_code = fsm.product_code  
    AND fgp.fiscal_year = fsm.fiscal_year  
JOIN dim_date dd  
ON dd.calender_date = fsm.date  
WHERE dd.fiscal_year IN (2020,2021)),
```



```
market_yearly_report AS (  
  SELECT gsr.market ,  
         SUM(CASE WHEN fiscal_year = 2020 THEN gross_sales ELSE 0 END)  
  AS fy_2020_sales,  
         SUM(CASE WHEN fiscal_year = 2021 THEN gross_sales ELSE 0 END)  
  AS fy_2021_sales  
  FROM gross_sales_report gsr  
  GROUP BY gsr.market  
)  
  
SELECT  
  myr.market,  
  ROUND(myr.fy_2020_sales/1000000,2) AS fy_2020_mln ,  
  ROUND(myr.fy_2021_sales/1000000,2) AS fy_2021_mln,  
  ROUND((myr.fy_2021_sales - myr.fy_2020_sales) /  
  NULLIF(myr.fy_2020_sales, 0) * 100,  
  2) AS grow_pct  
  FROM market_yearly_report myr  
  ORDER BY myr.market;
```



Results

DAY 11 / 15

06/08/26

fact_sales_monthly
dim_customer
fact_gross_price
SQL File 5*
dim_date

Limit to 50000 rows

```

15
16
17 market_yearly_report AS (
18     SELECT gsr.market ,
19     SUM(CASE WHEN fiscal_year = 2020 THEN gross_sales ELSE 0 END) AS fy_2020_sales,
20     SUM(CASE WHEN fiscal_year = 2021 THEN gross_sales ELSE 0 END) AS fy_2021_sales

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

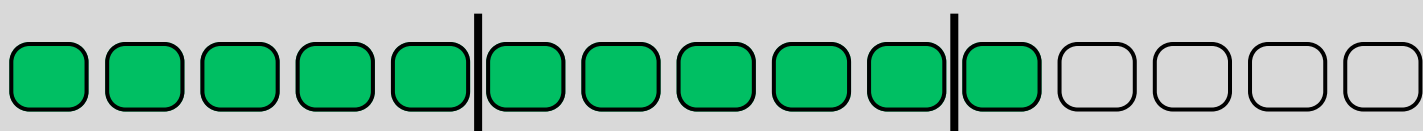
	market	fy_2020_mln	fy_2021_mln	grow_pct
▶	Australia	23.81	59.15	148.42
	Austria	0.31	8.25	2589.72
	Bangladesh	5.66	19.05	236.78
	Brazil	2.33	2.14	-8.11
	Canada	29.11	89.78	208.40
	Chile	0.19	1.46	664.20
	China	13.59	55.29	307.00
	Columbia	0.03	0.37	1062.99
	France	19.36	67.62	249.22
	Germany	13.74	41.25	200.16
	India	139.70	455.05	225.73
	Indonesia	14.85	48.12	224.12
	Italy	14.24	38.10	167.66
	Japan	4.91	17.34	253.19

Result 11

Output

Action Output

#	Time	Action	Message
✓ 1	16:52:59	WITH gross_sales_report AS (SELECT dc.customer_code , dc.customer , dc.channel, dc.market, dd.fiscal_year, fs...	27 row(s) returned



| Vishnu Ram Sachin | Data Analyst



DAY 11 / 15

06/08/26



I Vishnu Ram Sachin I Data Analyst

SQL THROUGH ONE DATASET CHALLENGE